

Stability analysis of networked control systems with denial-of-service attacks

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Abstract—This paper addresses stability problem for a class of networked control systems (NCSs) with network attacks. As a network attack to control systems, denial-of-service attacks are fully taken into account in the NCSs modelling. Under the scenario that there exist denial-of-service attacks in the communication channel, a closed-loop system with an error function is provided for the studied NCSs, where the denial-of-service attacks are well considered. Then based on the above constructed system model, one sufficient condition is derived for ensuring the uniformly ultimately boundedness of the derived system under consideration, and the upper bound of the maximum allowable number of denial-of-service attack is obtained. Moreover, the celebrated time-delay system technology method is introduced to improve the result of the upper bound of the maximum allowable number of denial-of-service attack. Finally, one numerical example is given to show the effectiveness of the proposed method.

Index Terms—Networked control systems, uniformly ultimately bounded, denial-of-service.

I. INTRODUCTION

The traditional communication architecture for control systems, which has been successfully implemented in industry for decades, is point-to-point. However, as the production processes are becoming more and more complex, the traditional point-to-point ways of connection have gradually reached their application limit [1]. With the proliferation of computer networks, common-bus network architectures are introduced to meet new requirements, such as quick and easy maintenance, integrated diagnostics and low cost through reduced wiring and distributed intelligence. It is called networked control system (NCS), where a feedback control system is communicated with sensors and actuators over a communication network [2], [3], [4].

It is well known that the insertion of the communication network in the feedback loop brings many issues. The first issue is network-induced delay that occurs while exchanging data among devices connected to the shared medium [5], [6], [7]. This delay, either constant or random, can degrade the performance of control system and even destabilize the system when the delay is not explicitly considered in the design process. The second issue is packet losses due to unreliable network transmission which can cause a loss in control input [8], [9]. During the past decades, the stability analysis and controller synthesis for NCSs have experienced an

increasing attention in the presence of network-induced delay and packet losses, see, for example [10], [11], [12], [13], [14] and reference therein.

Recently, with the development of the NCS, network, as a critical element in NCS, is vulnerable to cyber-threats which can menace the control system [15]. In the presence of network attack, the separation of the design stage of network security and control theory is no longer applicable. Therefore, the design for NCS under consideration of network attack is extremely urgent [16]. As a fact, a real attacker has no incentives to follow the traditional failure mode assumptions and may attack in a non-deterministic manner, which increases the difficulty of analysis and design of networked control systems. Notice that a common network attack style is so-called denial of service (DoS). DoS attack usually prevents the information exchange through a large volume invalid data to deliberately consume the network resources. However, up to now, little work has been found in the open literature on networked control systems with network DoS. For instant, for a class of DoS attack models, a feedback controller that minimize a given function subjected to safety and power constraints is designed in [17]. With an assumption that the attack behavior is a strategic jammer whose goal is to perturb the control system optimally, the saddle point equilibrium control jamming strategies has been analyzed by zero-sum method in [18]. For a signal-input remote control system, a control strategy with placing poles method is studied under the periodic DoS attack in [19]. Considering the scenario that DoS occurred during the data transmission between sensor and remote estimator with energy constraints, a optimal attack schedule that to maximize the expected average estimation error is studied in [20]. Notice that only the DoS schedule and strategic jammer are considered in the references mentioned above, and the stability problem of networked control systems in the presence of DoS attacks is not fully taken into account. This is the motivation of this work.

In this paper, we aim to analyze networked control systems in the presence of denial-of-service attacks. Considering characteristic of denial-of-service attacks that preventing the information exchange, a closed-loop system with an error function is presented for the studied networked control systems with DoS attacks. Then, by use of Lyapunov functional, one

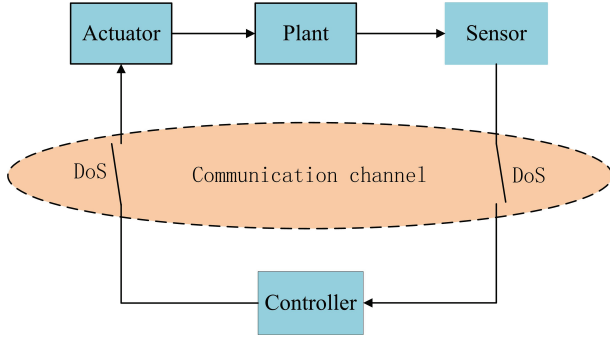


Fig. 1: Diagram for networked control systems with denial-of-service attack.

stability criterion is established for the derived closed-loop systems, which ensures that the studied networked control systems are said to be uniformly ultimately. Moreover, the upper bound of the maximum allowable number of denial-of-service attack bounded is given, and for a comparison, a less conservative result of the maximum allowable number of denial-of-service attack is provided by use of time-delay system technology method. Finally, one numerical example is given to show the effectiveness of the proposed method.

Notation: Throughout the paper, $\mathbb{N}$ denotes the natural numbers, $\mathbb{R}^n$ denotes the n -dimensional Euclidean space; $\mathbb{R}^{n \times m}$ is the set of $n \times m$ real matrices; I is the identity matrix of appropriate dimensions. For $X \in \mathbb{R}^{n \times n}$, notation $X > 0$ (respectively, $X \geq 0$) means that matrix X is a real symmetric positive definite (respectively, positive semi-definite).

II. SYSTEM DESCRIPTION AND PRELIMINARIES

Consider the following discrete-time system

$$x(k+1) = Ax(k) + Bu(k) \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the state vector, $u(k) \in \mathbb{R}^m$ is the control input vector, A and B are constant matrices with appropriate dimensions. It is naturally assumed that the pair (A, B) is controllable and there exists a state feedback matrix K such that all the eigenvalues of $A + BK$ have negative real parts.

It is well known that there are two types of malicious cyber attacks to control systems, i.e., deception attacks and DoS attacks [17]. In addition, as shown by the analysis of a database that tracked cyber-incidents affecting industrial control systems from 1982 to 2003, DoS attack is the most likely threat to control systems [21]. Therefore, we focus on DoS attacks to control systems as shown in Fig. 1 and leave deception attacks for future work.

Before proceeding further, it is assumed that the sensor measures the plant's state $x(k)$ at each sampling instant k th ($k \in \{0, 1, 2, \dots\}$), and sends it to the remote controller via a communication channel. Once the packet arrives at the controller, the controller immediately calculates and generates a new control signal, then the control packet is finally transmitted by the communication channel to the actuator with a zero-order-holder (ZOH). The role of ZOH is to keep the actuator's

control signal constant when no newest arrived control signal arrives. And the control signal will arrive at the remote actuator if there is no DoS attack during the transmission.

Under consideration of DoS attacks where the controller is prevented from receiving sensor measurement or the actuator is prevented from receiving control signals, the set of successful arrived at the actuator instant is denoted by $\{s_0, s_1, s_2, \dots\}$, where $s_0 = 0$ is the initial arrived instant. In what follows, for each $k \in \mathbb{N}$, the control input applied to the process can be expressed as

$$u(k) = Kx(s_i), \quad k \in [s_i, s_{i+1}) \quad (2)$$

where s_i represents the last successful control update, i.e. $s_i = \operatorname{argmin}_{s_j} \{k - s_j | k > s_j, j \in \mathbb{N}\}$.

Remark 1: It is clear that $\{s_0, s_1, s_2, \dots\}$ is a subset of the sampled instant set $\{0, 1, 2, \dots\}$. Notice that if there is no DoS attack, $\{s_0, s_1, s_2, \dots\}$ equals to $\{0, 1, 2, \dots\}$, that is, all the sampled signals are successful arriving at the actuator, which is reduced to a traditional control system case.

Consider the closed-loop system resulting from (1) under a control signal as in (2). First, we rewrite (2) in a new form that is better suited for analysis purposes. Denote

$$e(k) \triangleq x(s_i) - x(k), \quad k \in [s_i, s_{i+1}) \quad (3)$$

where $e(k)$ represents the error between the value of the process state at the last successful control update and the value of the process state at the current time. Therefore, the closed-loop system can be expressed as

$$x(k+1) = \Phi x(k) + BK e(k), \quad k \in [s_i, s_{i+1}) \quad (4)$$

where $\Phi \triangleq A + BK$.

For system (4) we consider the following definition borrowed from [22], which is used in the derivation of the main results of this work.

Definition 1: System (4) is said to be uniformly ultimately bounded in the set φ if for every initial condition $x_0 = x(0)$ there exists $T(x_0)$ such that for a given $e(k)$ we have $x(k) \in \varphi$ for $k > T(x_0)$.

Remark 2: Notice that by some elaborate transformations, the closed-loop system (4) now depends on the control update rule through $e(k)$. It can be intuitively seen that the uniformly ultimately boundedness of system (4) cannot be guaranteed with a non-ideal control update rule. Therefore, it is worth adopting a reasonable control update rule that keeps $e(k)$ small in a suitable sense, which will be analyzed in the following section.

III. MAIN RESULTS

In this section, we will firstly provide a suitable control update rule for guaranteeing the uniformly ultimately boundedness of system (4) by use of the Lyapunov functional approach.

Theorem 1: For a given matrix K , if there exist matrices $P > 0$ and $N > 0$ such that the following matrix inequality

$$\Phi^T P \Phi + \Phi^T P N^{-1} P \Phi - P < 0 \quad (5)$$

holds, then the states of system (4) are uniformly ultimately bounded and results in

$$\|x(k)\| \leq \sqrt{\frac{\epsilon\beta}{\alpha\lambda_{\min}\{P\}}} \quad (6)$$

where $\alpha = \frac{\lambda_{\min}\{-Q\}}{\lambda_{\max}\{P\}}$, $\beta = \lambda_{\max}\{K^T B^T (P + N) BK\}$ and ϵ is a small positive constant, if

$$\|A^{k-s_i} + \sum_{j=0}^{k-s_i-1} A^j BK - I\| \|x(s_i)\| < \sqrt{\epsilon}. \quad (7)$$

Proof 1: Choose a Lyapunov function candidate as

$$V(k) = x^T(k)Px(k) \quad (8)$$

with $P > 0$ being a solution of (4).

Defining $\Delta V(k) := V(k+1) - V(k)$ for $k \in [s_i, s_{i+1})$, we obtain

$$\begin{aligned} \Delta V(k) &= x^T(k+1)Px(k+1) - x^T(k)Px(k) \\ &\leq x^T(k)(\Phi^T P \Phi + \Phi^T P N^{-1} P \Phi - P)x(k) \\ &\quad + e^T(k)K^T B^T (P + N)BK e(k) \\ &\leq -\lambda_{\min}\{-Q\}x^T(k)x(k) \\ &\quad + \lambda_{\max}\{K^T B^T (P + N)BK\}\|e(k)\|^2 \\ &\leq -\frac{\lambda_{\min}\{-Q\}}{\lambda_{\max}\{P\}}x^T(k)Px(k) \\ &\quad + \lambda_{\max}\{K^T B^T (P + N)BK\}\|e(k)\|^2 \\ &\leq -\alpha V(k) + \beta \|e(k)\|^2 \end{aligned} \quad (9)$$

where $-Q = \Phi^T P \Phi + \Phi^T P N^{-1} P \Phi - P$. According to (9), it is easy to show that the following inequality holds for $k = s_i, s_i + 1, \dots, s_{i+1} - 2$

$$V(k) \leq (1 - \alpha)^{k-s_i} V(s_i) + \sum_{j=1}^{k-s_i} (1 - \alpha)^{j-1} \beta \|e(k-j)\|^2 \quad (10)$$

When $k = s_{i+1} - 1$, it follows from (9) that

$$\begin{aligned} V(s_{i+1}) &\leq (1 - \alpha)V(s_{i+1} - 1) + \beta \|e(s_{i+1})\|^2 \\ &\vdots \\ &\leq (1 - \alpha)^{s_{i+1}-s_i} V(s_i) \\ &\quad + \sum_{j=1}^{s_{i+1}-s_i} (1 - \alpha)^{j-1} \beta \|e(s_{i+1}-j)\|^2 \end{aligned} \quad (11)$$

Therefore it can be concluded that (10) holds for $k = [s_i, s_{i+1})$. Then, it follows from (10) and (11) that

$$\begin{aligned} V(k) &\leq (1 - \alpha)^{k-s_i} V(s_i) + \sum_{j=1}^{k-s_i} (1 - \alpha)^{j-1} \beta \|e(k-j)\|^2 \\ &\vdots \\ &\leq (1 - \alpha)^k V(0) + \sum_{j=1}^k (1 - \alpha)^{j-1} \beta \|e(k-j)\|^2 \\ &\leq (1 - \alpha)^k \lambda_{\max}\{P\} \|x(0)\|^2 \\ &\quad + \sum_{j=1}^k (1 - \alpha)^{j-1} \beta \|e(k-j)\|^2 \end{aligned} \quad (12)$$

Notice that

$$V(k) \geq \lambda_{\min}\{P\} \|x(k)\|^2 \quad (13)$$

which yields

$$\begin{aligned} \|x(k)\|^2 &\leq (1 - \alpha)^k \frac{\lambda_{\max}\{P\}}{\lambda_{\min}\{P\}} \|x(0)\|^2 \\ &\quad + \sum_{j=1}^k (1 - \alpha)^{j-1} \frac{\beta}{\lambda_{\min}\{P\}} \|e(k-j)\|^2 \end{aligned} \quad (14)$$

Moreover, from the definition of the error $e(k)$, we obtain that for $k = s_i + 1, s_i + 2, \dots, s_{i+1} - 1$

$$\begin{aligned} e(k) &= -Ax(k-1) - (BK - I)x(s_i) \\ &\vdots \\ &= -(A^{k-s_i} + \sum_{j=0}^{k-s_i-1} A^j BK - I)x(s_i) \end{aligned} \quad (15)$$

and $e(k) = 0$ if $k = s_i$. It follows from (15) that

$$\|e(k)\| \leq \|A^{k-s_i} + \sum_{j=0}^{k-s_i-1} A^j BK - I\| \|x(s_i)\| \quad (16)$$

Substituting (7) into (16) yields

$$\|e(k)\| < \sqrt{\epsilon}, \quad k = [s_i, s_{i+1}) \quad (17)$$

From (14) and (17), it implies that the states of the system (4) converge into the bounded region and converge to the bounded region

$$\|x(k)\| \leq \sqrt{\frac{\epsilon\beta}{\alpha\lambda_{\min}\{P\}}} \quad (18)$$

thus by use of Definition 1, the uniformly ultimately boundedness of the system (4) can be guaranteed. This completes the proof. $\square$

Remark 3: Notice that a similar boundedness criterion has been provided in [23] by directly imposing on the error $e(k)$. The difference here lies in the bound, which indirectly imposes on the left side of the inequality (7) including $\|A^{k-s_i} + \sum_{j=0}^{k-s_i-1} A^j BK - I\|$ and $\|x(s_i)\|$. By use of

the condition (16), we can also restrict the error $e(k)$ that ensures the uniformly ultimately boundedness of the system (4) without prior knowledge of an upper bound on $e(k)$.

In what follows, one needs to constrain the maximum allowable number of denial-of-service attack d_{MANDoSA} , namely the length of the intervals over the communication channel is interrupted. For this purpose, we provide the following result on the basis of Theorem 1.

Theorem 2: From (7), the maximum allowable number of denial-of-service attack d_{MANDoSA} is upper bounded by

$$d_i \leq d_{\text{MANDoSA}} \triangleq \text{Floor} \left(\frac{\sqrt{\epsilon} + \sqrt{\epsilon} \|A - I\|}{\|\Phi - I\| \|x(s_i)\| + \sqrt{\epsilon} \|A - I\|} \right) \quad (19)$$

where $\text{Floor}(\diamond)$ equals to the integer part for $\diamond \in \mathbb{R}$.

Proof 2: Consider a successful broadcast release interval $[s_i, s_{i+1})$. For $k = s_i, s_i + 1, \dots, s_{i+1} - 1$, we obtain

$$\begin{aligned} \|x(k+1) - x(k)\| &\leq \|(\Phi - I)x(k) + BKe(k)\| \\ &= \|(\Phi - I)x(s_i) + (I - A)e(k)\| \\ &\leq \|\Phi - I\| \|x(s_i)\| + \|A - I\| \|e(k)\| \end{aligned} \quad (20)$$

Combining (17) and (20) together leads to

$$\begin{aligned} \|x(k+1) - x(k)\| &\leq \begin{cases} \|\Phi - I\| \|x(s_i)\|, & k = s_i \\ \|\Phi - I\| \|x(s_i)\| + \sqrt{\epsilon} \|A - I\|, & \text{otherwise} \end{cases} \end{aligned} \quad (21)$$

Assuming that in the interval $[s_i, s_{i+1})$, there are d_i unsuccessfully transmitted broadcast packets, i.e. $d_i \triangleq s_{i+1} - s_i - 1 \in \mathbb{N}$, one can obtain

$$\|x(s_i + d_i) - x(s_i)\| \leq \sum_{k=s_i}^{s_i+d_i-1} \|x(k+1) - x(k)\| \quad (22)$$

Then it follows from (21) and (22) that

$$\begin{aligned} \|x(s_i + d_i) - x(s_i)\| &\leq d_i \|\Phi - I\| \|x(s_i)\| + (d_i - 1) \sqrt{\epsilon} \|A - I\| \end{aligned} \quad (23)$$

From (23), one can see that the condition (7) in Theorem 1 is ensured by (19), which implies that Theorem 2 can be readily derived from Theorem 1. This completes the proof. $\square$

In Theorem 2, the existence of the upper bound of the maximum allowable number of denial-of-service attack d_{MANDoSA} has been shown depending on the norm of $x(s_i)$. Actually, for each instant, the upper bound of d_{MANDoSA} can be determined analytically which is independent on $x(s_i)$. For example, defining an ‘‘artificial delay’’ $\eta(k) = k - s_i$ for $k \in [s_i, s_{i+1})$, it follows from (1) and (2) that

$$x(k+1) = Ax(k) + BKx(k - \eta(k)) \quad (24)$$

and $\eta(k)$ belongs to the interval $[0, s_{i+1} - s_i)$, e.g. $\eta(k) \in [0, d_{\text{MANDoSA}}]$. Then, by use of the lower and upper bounds of $\eta(k)$, one constructs a Lyapunov functional candidate as

$$\begin{aligned} V(k) &= x^T(k)Px(k) + \sum_{j=k-d_{\text{MANDoSA}}}^{k-1} x^T(j)Mx(j) \\ &\quad + d_{\text{MANDoSA}} \sum_{\theta=-d_{\text{MANDoSA}}}^{-1} \sum_{j=k+\theta}^{k-1} y^T(j)Ny(j) \end{aligned} \quad (25)$$

with $y(k) = x(k+1) - x(k)$, $P > 0$, $M > 0$ and $N > 0$. Based on the notable reciprocally convex approach [24], one can readily obtain a stability criterion. Since the similar results on stability of time delay system can be seen in [25], the detail criterion is omitted in this paper.

Remark 4: Compared with the proposed Lyapunov functional (8), two additional Lyapunov functionals are constructed in (25), which includes the lower and upper bounds of the delay $\eta(k)$. It is worth pointing out that since the two Lyapunov functionals are additionally constructed, a significant improvement on the maximum allowable number of denial-of-service attack d_{MANDoSA} can be expected, which will be theoretically proved in the following section.

Remark 5: Notice that in [23], the lower bound of the DoS duration should be specially analyzed in order to ensure that no Zeno behavior occurs. However, in this paper, one can see that the DoS duration is at least lower bounded by the constant 1. It implies that Zeno behavior is absolutely excluded, thus there is no essential to analyze the lower bound of the DoS duration.

Remark 6: Only the DoS duration has been studied, and the other characteristic of DoS attacks, e.g. DoS frequency, is not investigated in this paper. It is clear that the frequency at which DoS occurs cannot be much huge which may lead to instability of control system. However, it is not difficult to further extend the above mentioned results to the problem of DoS frequency. For example, one can easily obtain the equivalent switched systems for the closed-loop system (4); then by using the concept of average dwell-time, a bound of DoS frequency can be designed to guarantee that system (4) is stable, which is a related direction for the future research.

IV. NUMERICAL EXAMPLE

We now illustrate the previous results on the linear control system [25]

$$\begin{aligned} \begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} &= \begin{bmatrix} 1.0078 & 0.0301 \\ 0.5202 & 1.0078 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} \\ &\quad + \begin{bmatrix} -0.0001 \\ -0.0053 \end{bmatrix} u(k) \end{aligned} \quad (26)$$

and a linear feedback $u(k) = 101.9472x_1(s_i) + 24.5307x_2(s_i)$.

By use of Theorem 1 and the Matlab LMI Toolbox, we obtain a feasible result, i.e.,

$$P = \begin{bmatrix} 5.6742 & 9.1948 \\ 9.1948 & 58.4792 \end{bmatrix}, \quad N = \begin{bmatrix} 346.4385 & 29.9356 \\ 29.9356 & 326.1032 \end{bmatrix}$$

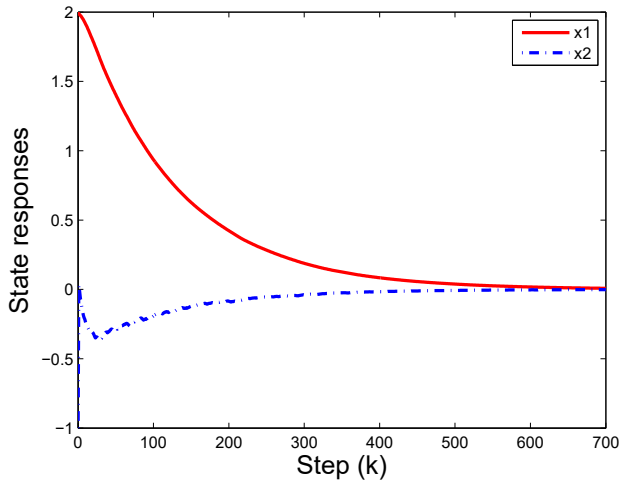


Fig. 2: State responses of networked control systems with denial-of-service attack.

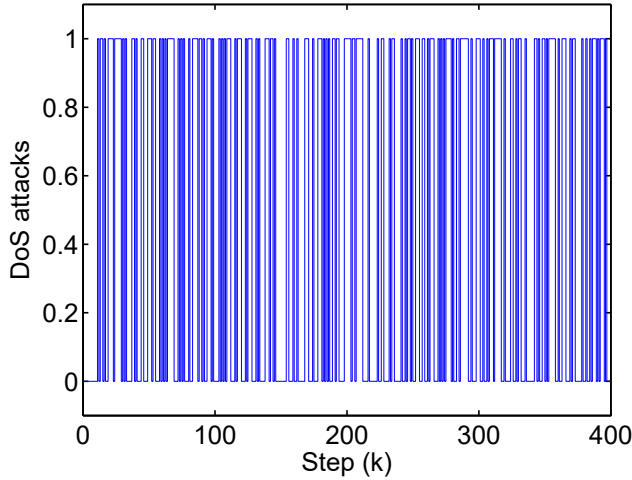


Fig. 3: Stochastic denial-of-service attacks.

which shows that the given controller gain ensures the uniformly ultimately boundedness of the system (4). Moreover, with a given parameter $\varepsilon = 10^{-6}$, one can see that the states of the system (4) can converge into the bounded region and converge to the bounded region $\|x(k)\| \leq 0.5896$.

From Theorem 2, one obtains d_{MANDoSA} is 1 and 2 if $\|x(s_i)\|$ equals to 0.004 and 0.001, respectively. As is mentioned above, this result is much conservative and depends on the value of the system state $x(s_i)$. To improved this result, we adopt the time delay system method, and obtain $d_{\text{MANDoSA}} = 7$, which shows the effectiveness of the proposed method.

Furthermore, applying the above parameters, the state responses of the closed-loop system are shown in Fig. 2 under a given initial condition as $x(0) = [2, -1]^T$ and DoS attacks

shown in Fig. 3 where 1 implies that there is a DoS attack jamming to control system at the k th instant. It can be seen from the curves in Fig. 2 that the proposed method is effective, which ensures that the resulting closed-loop systems are the uniformly ultimately bounded with DoS attacks.

V. CONCLUSION

Under consideration of the effect of denial-of-service attacks, a closed-loop system has been proposed for the studied networked control systems. It is worth mentioned that an error function has been introduced in the derived closed-loop system to express the error between the value of the process state at the last successful control update and the value of the process state at the current time. By use of Lyapunov functional method, one sufficient condition has been obtained for ensuring the uniformly ultimately boundedness of the derived system under consideration. Then the upper bound of the maximum allowable number of denial-of-service attack has been also obtained based on the derived sufficient condition. Moreover, the celebrated time-delay system technology method has been introduced to improve the result of the upper bound of the maximum allowable number of denial-of-service attack. Finally, one numerical example has been given to show the effectiveness of the proposed method.

ACKNOWLEDGMENT

This work was supported in part by the National Natural Science Foundation of China under Grant 61273114, the Innovation Program of Shanghai Municipal Education Commission under Grant 14ZZ087, the Pujiang Talent Plan of Shanghai City, China under Grant 14PJ1403800, the International Corporation Project of Shanghai Science and Technology Commission under Grant 14510722500, the Program for Professor of Special Appointment (Eastern Scholar) at Shanghai Institutions of Higher Learning, and the Natural Science Foundation of Jiangsu Province of China under Grant BK20131403.

REFERENCES

- [1] T. C. Yang, "Networked control system: a brief survey," *IEE Proceedings Control Theory and Applications*, vol. 153, no. 4, p. 403, 2006.
- [2] W. Zhang, M. S. Branicky, and S. M. Phillips, "Stability of networked control systems," *IEEE Control Systems*, vol. 21, no. 1, pp. 84–99, 2001.
- [3] C. Peng and J. Zhang, "Event-triggered output-feedback H_∞ control for networked control systems with time-varying sampling," *IET Control Theory & Applications*, vol. 9, no. 9, pp. 1384–1391, 2015.
- [4] R. A. Gupta and M.-Y. Chow, "Networked control system: overview and research trends," *IEEE Transactions on Industrial Electronics*, vol. 57, no. 7, pp. 2527–2535, 2010.
- [5] L. Zhang, Y. Shi, T. Chen, and B. Huang, "A new method for stabilization of networked control systems with random delays," *IEEE Transactions on Automatic Control*, vol. 50, no. 8, pp. 1177–1181, 2005.
- [6] J. Zhang and C. Peng, "Event-triggered H_∞ filtering for networked takagi–sugeno fuzzy systems with asynchronous constraints," *IET Signal Processing*, vol. 9, no. 5, pp. 403–411, 2015.
- [7] X. Ge, F. Yang and Q.-L. Han. Distributed networked control systems: a brief overview," *Information Sciences*, 2015, doi: 10.1016/j.ins.2015.07.047.
- [8] W.-A. Zhang and L. Yu, "Output feedback stabilization of networked control systems with packet dropouts," *IEEE Transactions on Automatic Control*, vol. 52, no. 9, pp. 1705–1710, 2007.

- [9] C. Peng, E. Tian, J. Zhang, and D. Du, "Decentralized event-triggering communication scheme for large-scale systems under network environments," *Information Sciences*, 2015, doi: 10.1016/j.ins.2015.06.036.
- [10] J. Zhang, C. Peng, M.-R. Fei, and Y.-C. Tian, "Output feedback control of networked systems with a stochastic communication protocol," *Journal of the Franklin Institute*, 2016, doi: 10.1016/j.jfranklin.2016.02.009.
- [11] W.-A. Zhang, L. Yu, and S. Yin, "A switched system approach to H_∞ control of networked control systems with time-varying delays," *Journal of the Franklin Institute*, vol. 348, no. 2, pp. 165–178, 2011.
- [12] C. Peng and Q.-L. Han, "On designing a novel self-triggered sampling scheme for networked control systems with data losses and communication delays," *IEEE Transactions on Industrial Electronics*, vol. 63, no. 2, pp. 1239–1248, 2016.
- [13] S. Hu, D. Yue, X. Xie, and Z. Du, "Event-triggered H_∞ stabilization for networked stochastic systems with multiplicative noise and network-induced delays," *Information Science*, vol. 299, pp. 178–197, 2015.
- [14] J. Zhang, C. Peng, D. Du, and M. Zheng, "Adaptive event-triggered communication scheme for networked control systems with randomly occurring nonlinearities and uncertainties," *Neurocomputing*, vol. 174, pp. 475–482, 2016.
- [15] H. Sandberg, S. Amin, and K. Johansson, "Cyberphysical security in networked control systems: an introduction to the issue," *Control Systems, IEEE*, vol. 35, no. 1, pp. 20–23, 2015.
- [16] A. Cardenas, S. Amin, B. Sinopoli, A. Giani, A. Perrig, and S. Sastry, "Challenges for securing cyber physical systems," in *Workshop on future directions in cyber-physical systems security*, p. 5, 2009.
- [17] S. Amin, A. A. Cárdenas, and S. S. Sastry, "Safe and secure networked control systems under denial-of-service attacks," in *Hybrid Systems: Computation and Control*, pp. 31–45, Springer, 2009.
- [18] A. Gupta, C. Langbort, and T. Basar, "Optimal control in the presence of an intelligent jammer with limited actions," in *49th IEEE Conference on Decision and Control*, pp. 1096–1101, IEEE, 2010.
- [19] H. S. Foroush and S. Martínez, "On single-input controllable linear systems under periodic DoS jamming attacks," *arXiv preprint arXiv:1209.4101*, 2012.
- [20] H. Zhang, P. Cheng, L. Shi, and J. Chen, "Optimal denial-of-service attack scheduling with energy constraint," *IEEE Transactions on Automatic Control*, vol. 60, no. 11, pp. 3023–3028, 2015.
- [21] E. Byres and J. Lowe, "The myths and facts behind cyber security risks for industrial control systems," in *Proceedings of the VDE Kongress*, vol. 116, pp. 213–218, 2004.
- [22] F. Blanchini, "Ultimate boundedness control for uncertain discrete-time systems via set-induced lyapunov functions," in *Proceedings of the 30th IEEE Conference on Decision and Control*, pp. 1755–1760, IEEE, 1991.
- [23] C. De Persis and P. Tesi, "Input-to-state stabilizing control under denial-of-service," *IEEE Transactions on Automatic Control*, vol. 60, no. 11, pp. 2930–2944, 2015.
- [24] P. Park, J. W. Ko, and C. Jeong, "Reciprocally convex approach to stability of systems with time-varying delays," *Automatica*, vol. 47, no. 1, pp. 235–238, 2011.
- [25] J. Zhang, C. Peng, and M. Zheng, "Improved results for linear discrete-time systems with an interval time-varying input delay," *International Journal of Systems Science*, vol. 47, no. 2, pp. 492–499, 2016.